

HTTP Status Codes — Poora Guide

Backend development me use hone wale sabhi status codes, unka matlab aur kaam

HTTP status code ek 3-digit number hota hai jo server, client (browser/app) ko batata hai ki unki request ka kya result raha — success mila, redirect hua, client ki galti thi, ya server me hi kuch kharab hua. Pehla digit category batata hai:

1xx	Informational — request receive hui, processing chal rahi hai
2xx	Success — request successfully complete hui
3xx	Redirection — aage kisi aur jagah jaana hai
4xx	Client Error — request bhejne wale (client) ki galti
5xx	Server Error — server side ki galti/problem

1xx — Informational Responses

Code	Naam (Name)	Kya kaam karta hai (Explanation)
100	Continue	Server ne request ka pehla part receive kar liya hai, client baaki request bhejta rahe. Bade file upload me use hota hai.
101	Switching Protocols	Server client ki request maan kar protocol switch kar raha hai (jaise HTTP se WebSocket).
102	Processing	Server request process kar raha hai lekin response abhi ready nahi hai (WebDAV me use).
103	Early Hints	Final response se pehle server kuch resource hints bhej deta hai taaki browser preload kar sake.

2xx — Success Responses

Code	Naam (Name)	Kya kaam karta hai (Explanation)
200	OK	Sabse common success status. Request successfully process ho gayi aur response bhi mil gaya.
201	Created	Naya resource successfully create ho gaya, jaise POST request se database me naya record ban gaya.
202	Accepted	Request accept ho gayi hai lekin processing abhi complete nahi hui, background me chal rahi hai.
203	Non-Authoritative Info	Response mila hai lekin original server se nahi, kisi proxy/cache se aaya hai.
204	No Content	Request successful hai lekin response body me kuch bhejna nahi hai, jaise DELETE ke baad.
205	Reset Content	Server चाहता है client apna form/view reset kar de.
206	Partial Content	Sirf resource ka ek part bheja gaya hai, jaise video streaming ya resumable download me.

3xx — Redirection Responses

Code	Naam (Name)	Kya kaam karta hai (Explanation)
300	Multiple Choices	Ek se zyada response options available hain, client ko choose karna hai.
301	Moved Permanently	URL permanently doosre URL par shift ho gaya hai, browser/search engine naya URL yaad rakhte hain.
302	Found	URL temporarily doosri jagah redirect ho raha hai, original URL future me kaam karega.
303	See Other	Response doosre URL par GET request se lena chahiye (POST ke baad common).
304	Not Modified	Cached version already up-to-date hai, server dobara data nahi bhejta, bandwidth bachta hai.
307	Temporary Redirect	302 jaisa hi hai lekin method (POST/GET) same rehta hai redirect ke baad bhi.
308	Permanent Redirect	301 jaisa hi hai lekin request method change nahi hota.

4xx — Client Error Responses

Code	Naam (Name)	Kya kaam karta hai (Explanation)
400	Bad Request	Client ne galat/invalid syntax wali request bheji hai, server samajh nahi paaya.
401	Unauthorized	Authentication required hai - login/token missing ya galat hai.
402	Payment Required	Rarely use hota hai, payment related access control ke liye reserved hai.
403	Forbidden	Server request ko samajh gaya lekin permission nahi hai access karne ki (authorization fail).
404	Not Found	Requested resource/URL server par exist nahi karta.
405	Method Not Allowed	Jo HTTP method use kiya (GET/POST/PUT) wo us URL par allowed nahi hai.
406	Not Acceptable	Server client ke Accept header ke format me response nahi de sakta.
407	Proxy Auth Required	Pehle proxy server se authenticate hona padega.
408	Request Timeout	Client ne time par complete request nahi bheji, server ne connection close kar diya.
409	Conflict	Request current server state se conflict kar rahi hai, jaise duplicate entry.
410	Gone	Resource permanently delete ho chuka hai aur wapas nahi aayega (404 se zyada specific).
411	Length Required	Request me Content-Length header missing hai jo required tha.
412	Precondition Failed	Request headers me diya gaya condition server par match nahi hua.
413	Payload Too Large	Request body allowed size limit se bada hai.
414	URI Too Long	Request URL server ki limit se zyada lamba hai.
415	Unsupported Media Type	Request ka format (Content-Type) server support nahi karta, jaise XML bheja par JSON expect tha.
416	Range Not Satisfiable	Client ne jo byte-range maanga wo file me exist nahi karta.
417	Expectation Failed	Expect header me diya gaya condition server fulfill nahi kar saka.
422	Unprocessable Entity	Syntax sahi hai lekin data validation fail ho gaya, jaise required field missing (APIs me bahut common).
423	Locked	Resource currently locked hai, edit nahi ho sakta.
424	Failed Dependency	Ye request fail hui kyunki koi previous dependent request fail hui thi.
428	Precondition Required	Server chahta hai request conditional ho taaki conflicts avoid ho.
429	Too Many Requests	Rate limiting - client ne bahut zyada requests bhej di hain thode time me.
431	Header Fields Too Large	Request headers allowed size se bade hain.

Code	Naam (Name)	Kya kaam karta hai (Explanation)
451	Unavailable For Legal Reasons	Legal/government order ki wajah se content block kiya gaya hai.

5xx — Server Error Responses

Code	Naam (Name)	Kya kaam karta hai (Explanation)
500	Internal Server Error	Server ke andar hi kuch generic error/exception aa gayi, code me bug ho sakta hai.
501	Not Implemented	Server us functionality/method ko support hi nahi karta.
502	Bad Gateway	Server proxy/gateway ke roop me kaam kar raha tha, lekin upstream server se invalid response mila.
503	Service Unavailable	Server abhi overloaded hai ya maintenance me hai, temporarily available nahi.
504	Gateway Timeout	Gateway/proxy ne upstream server se time par response nahi mila, timeout ho gaya.
505	HTTP Version Not Supported	Client ne jo HTTP version use kiya wo server support nahi karta.
507	Insufficient Storage	Server ke paas request complete karne ke liye storage nahi bacha (WebDAV).
508	Loop Detected	Request process karte waqt infinite loop detect hua.
510	Not Extended	Request ko complete karne ke liye further extensions chahiye.
511	Network Auth Required	Client ko network access ke liye pehle authenticate karna padega (jaise WiFi login page).

Backend Code (Express.js) me res.status() — Practical Usage

Ye section specially backend code likhte waqt kaam aayega — jab tu khud `res.status(XXX).json()` likh raha hota hai to kaunsa status code kab dena chahiye, uska quick reference.

Example

```
notes.push(req.body);  
res.status(201).json({ message: "Note created", data: req.body });
```

Yaha 201 isliye use hua kyunki naya resource (note) create ho raha hai (push = naya data add ho raha hai array/DB me).

Sabse zyada use hone wale codes jo khud apne code me likhte hain:

Code	Kab use karein	Code Example
200	Normal success response — GET request se data fetch karke bhejte waqt.	<code>res.status(200).json(notes);</code>
201	Naya resource create hone ke baad — POST request me.	<code>res.status(201).json(newNote);</code>
204	Success hua par response body me kuch nahi bhejna — DELETE ke baad.	<code>res.status(204).send();</code>
400	Client ne galat data bheja — validation fail.	<code>res.status(400).json({error: "Title required"});</code>
401	User login nahi hai / token missing.	<code>res.status(401).json({error: "Please login"});</code>
403	User login toh hai par permission nahi (jaise admin-only route pe normal user).	<code>res.status(403).json({error: "Access denied"});</code>
404	Jo item/route dhoond raha hai wo exist nahi karta.	<code>res.status(404).json({error: "Note not found"});</code>
409	Duplicate entry — jaise same email se dobara signup.	<code>res.status(409).json({error: "Email exists"});</code>
422	Data ka format sahi hai par validation rules fail — jaise password chhota hai.	<code>res.status(422).json({error: "Password too short"});</code>
500	Server/code me hi error aa gaya (try-catch ke catch block me).	<code>res.status(500).json({error: err.message});</code>

Ek chhota real pattern jo har route me dikhega:

```
// GET all  
res.status(200).json(notes); // ■ mil gaya  
  
// POST new  
res.status(201).json(newNote); // ■ ban gaya  
  
// PUT/PATCH update  
res.status(200).json(updatedNote); // ■ update ho gaya  
  
// DELETE  
res.status(204).send(); // ■ delete ho gaya, kuch bhejna nahi
```

```
// Not found
res.status(404).json({error: "Not found"}); // ■ nahi mila

// Server crash / catch block
res.status(500).json({error: err.message}); // ■■ code me error
```

Basic Rule Yaad Rakhna: 2xx = sab thik | 4xx = galti request bhejne wale (client/frontend) ki | 5xx = galti apne backend code/server ki